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SIMULTANEOUS OPTIMIZATION OF HORIZONTAL AND VERTICAL ALIGNMENTS FOR HIGHWAYS

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Abstract—This paper presents the formulation of a three-dimensional highway alignment optimization model. The coordinates of the highway is parametrized by using a three-dimensional cubic spline polynomial interpolation. The solution is obtained by locating the optimal horizontal and vertical alignments simultaneously while subject to a wide range of possible constraints. These constraints may be bounds on the slope or curvature, as well as avoidance of inaccessible region. It is an extension from a previous research work where optimal vertical highway alignment was determined for a preselected horizontal alignment. Two numerical examples based upon the actual ground conditions in Singapore are presented to illustrate the applicability of the proposed model. Some computational details such as scaling and selection of initial variables which affect the rate of convergence are also briefly discussed.

INTRODUCTION

In a route location analysis between any two highway termini, a designer is often faced with a complicated task of selecting a path that has the lowest costs while conforming to certain constraints. These constraints may be imposed by operational requirements of vertical gradient, horizontal and vertical curvatures, socioeconomic considerations, the presence of inaccessible regions, and other factors. Traditionally, route location analysis has relied heavily on human judgement and intuition. With the advent of sophisticated mathematical techniques and high-speed digital computers, designers were able to perform the tasks in a more systematic and rigorous manner.

Dynamic Programming has been used by several researchers (Antoniotti, 1969; O'Brien and Bennett, 1969; Trietsch, 1987) to give minimum construction and users' costs solutions to the vertical highway alignment problems. Recently, Goh *et al.* (1988) had also extended this approach in locating the optimal path for the vertical highway alignment problem. Because of the limitation in computer capacity, the effectiveness of dynamic programming in solving a three-dimensional simultaneous vertical and horizontal alignment problem has yet to be proven. In most cases, coarse grids have to be used and this can only produce approximate suboptimal solution. Nicholson *et al.* (1976) proposed an improvement by first adopting a Dynamic Programming model to search for a band of interests within a feasible region, and then refining using a second model based upon discrete variational calculus to arrive at an optimal alignment.

Parker (1977) also employed a two-stage approach to determine the optimal path. The area is partitioned into square zones and the center of each zone is assigned a value equal to the average terrain elevation of the zone. The approach first finds, by a linear programming regression model, a surface covering the area of interest such that the total earthwork cost is minimized. Only vertical gradient constraints are observed in this first stage. The construction cost resulted from differences between elevations of the selected surface and the original ground are computed for each square zone. The second stage is to locate the best path on the selected surface that has the minimum cost. In another two-stage method, Howard *et al.* (1968) proposed to use the Optimum Curvature Principle to determine the horizontal alignment first and the vertical alignment subsequently by a two-dimensional optimization technique.

Each of the above models employed two stages of optimization and search grids to locate an optimum path. We would like to point out, however, that the three-dimensional highway alignment problem is unlikely to be solved completely by any two-stage model,

even if the model in each stage utilizes very fine grid. In other words, any two-stage model for solving the three-dimensional highway problem can at best produce a suboptimal solution. The motivation to reexamine the problem by a direct approach is thus imminent.

This paper presents an alternative approach of solving a three-dimensional highway route selection problem where the optimal horizontal and vertical alignments are determined simultaneously. It is a sequel to an earlier study (Goh *et al.*, 1988) that determines the optimal vertical highway alignment using state parametrization approach. Instead of using search grids, it regards the highway profile as a continuous function approximated by a series of cubic spline polynomial functions. The optimal profile is then determined by using standard nonlinear constraint optimization techniques. The proposed direct approach, though inevitably will incur more complexity in the mathematical modelling, enables a true optimal solution to be obtained. In addition, the demand for memory storage and computational CPU time is also well within the available limit.

COST AND CONSTRAINT COMPONENTS IN HIGHWAY CONSTRUCTION

The major cost components that contributed to the highway route location are listed below (Stott, 1972):

(i) *Construction costs*

Drainage	10%
Earthworks	25% (about half of this is haul cost)
Pavement	30%
Land	5% (to 10%)
Others	30%

(ii) *Vehicle operating costs* (net present values discounted over 30 years)
30%–1,000% of construction costs

(iii) *Design and administrative costs*
About 5% of construction costs

(iv) *Maintenance costs* (net present values discounted over 30 years)
About 5% of construction costs

We would like to give a brief description on some of the costs and how they are considered in the formulation of the proposed model.

Earthwork costs

The costs associated with earthwork are the costs of cutting and filling. Cutting costs are incurred when the design road proifle is lower than the ground profile. Filling costs are those costs of constructing embankments when the road profile is above the ground level. These costs are basically functions of the volume of earth moved, although the unit rate of excavation may vary with the depth of the cut and the types of soil. Transportation costs for hauling soil from borrow pits to dumping sites are also involved. For simplicity, the earthwork costs are computed from the volume of cut and fill. The cross section of the cut and fill are depicted in Fig. 1. Three variables are used in computing earthwork volume. They are the roadbed width w, cut or embankment θ, and the difference between the computed vertical road alignment and the ground level h.

Pavement costs

Pavement costs are computed from pavement thickness and the unit costs of pavement materials used. It is likely for pavement costs to vary along the road profile since the soil type and strength are more often than not nonhomogeneous in nature. These costs can be modelled as a function of locations in accordance with soil type distribution within the area of interest.

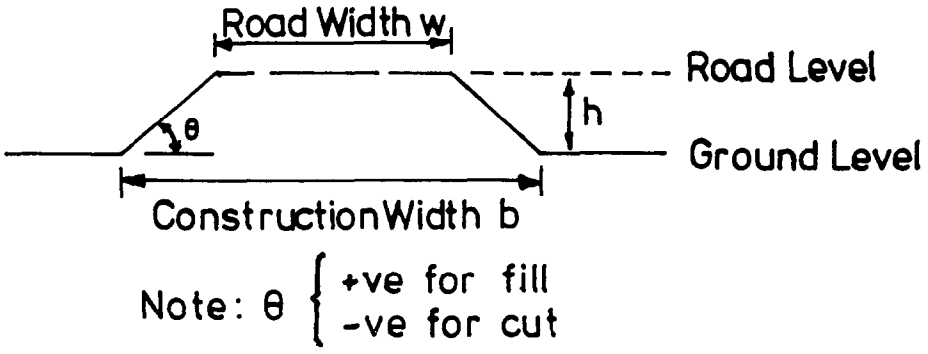

Fig. 1. Cross-sectional area of the cut and fill.

Vehicle operating costs

Vehicle operating costs include the costs of fuel, tires, engine oil, and maintenance of vehicles. To incorporate these cost items in the formulation, they can be expressed as functions of design speed, distance of travel, number of stops and speed changes, and vehicle types. These costs are not included in the present study due to lack of information on these relationships in Singapore.

Land acquisition costs

This is defined by the product of unit area land costs and the areas of land required for the highway right of way. The unit cost of land may vary along the length of the proposed route, depending on the type of construction or the nature of land development. This cost can therefore be handled in the same manner as pavement costs described earlier.

Drainage and embankment slope protection

Drainage facilities are generally not drastically affected by highway alignment, but are more dependent upon pavement material selected, climatic factors, and design of cross-sectional elements of the highway concerned.

The embankment slope protection cost is, however, affected by the embankment slope and material used; thus, it can be expressed as a function of embankment slope θ in Fig. 1 and pavement cost which is dependent on the type of construction material used.

The constraint components considered are vertical gradient bounds, horizontal and vertical curvature bounds, and inaccessible regions.

Vertical gradient bounds. Since restrictive sight distance is usually associated with steep uphill gradients, the speed of traffic is generally affected. As a result, not only the operating costs of vehicles increased but capacity of the road will have to be reduced to maintain the required level of service. Steep downhill gradients should also be avoided as accidents may occur due to increase in the length of the braking distance for the same design speed. Most highway authorities now accept a gradient of 4% as being the maximum desirable on major highways; hence, this criterion is usually a controlling factor in highway design.

Horizontal and vertical curvature bounds. When two different or contrary gradients meet, they are connected by a curve in the vertical plane known as the vertical curve. They are needed to secure gradual changes in grade so that abrupt changes in grade at the apex is avoided. This is also similar for horizontal curvature. The maximum allowable curvature is based upon the recommended practice of the American Association of State Highway Officials (AASHTO), which is generally dependent on the design speed as well as the sight distance. This is to prevent accidents due to inadequate visibility, discomfort, and damage due to humps and troughs.

Inaccessible regions. These may be bodies of waterways, protected military zones, or areas of poor soil condition that the highway must necessarily avoid. This constant

is explicitly considered in the model, which in the past had been neglected to the best of our knowledge.

CONTINUOUS STATE PARAMETRIZATION MODEL

The proposed model is formulated as a calculus of variation problem (Kamien and Schwartz, 1981). The road profile is represented by a continuous function in a Cartesian coordinate system as shown in Fig. 2. The x and y axes define the horizontal plane, and $f(x, y)$ and z represent the vertical coordinates of ground elevation and road profile, respectively. The surface ground profile is represented by a series of right angle planar triangular surfaces, each formed by three known vertical coordinates adjacent to each other. The smaller the triangle, i.e. the closer the coordinates are to each other, the closer the approximation to the true ground profile. The highway's coordinates are expressed as parametric equations of a parameter t where t is normalized to lie between 0 and 1. Like all constrained optimization problem, the objective function as the total cost may be expressed in a general form:

$$\text{Min } J = \int_0^1 \sum_{i=1}^{n} \phi_i(t, \mathbf{v}(\mathbf{t}), \mathbf{v}'(\mathbf{t}))\, \mathrm{d}t \tag{1}$$

subject to inequality constraints:

$$g_k(t, \mathbf{v}(\mathbf{t}), \mathbf{v}'(\mathbf{t}), \mathbf{v}''(\mathbf{t})) \leq C_k \quad k = 1, \ldots, m, \tag{2}$$

where $\mathbf{v}$ is a vector $(x, y, z)^T$, representing the location of the highway axes and $t \in [0, 1]$; ϕ_i, $i = 1, \ldots, n$ are the n cost components under consideration; m is the number

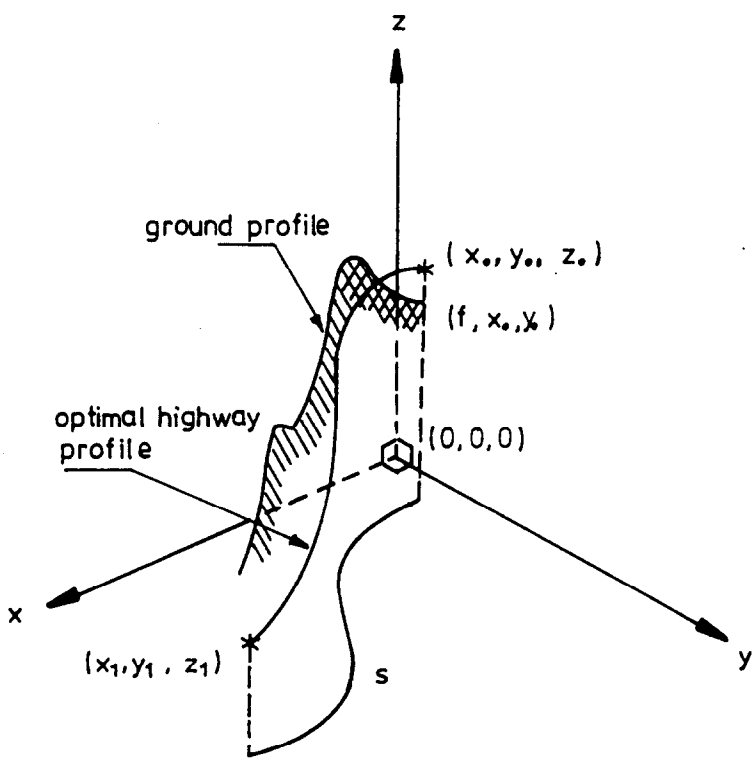

Fig. 2. Cartesian coordinate system representation of the ground and road profiles.

of constraints involved. The fixed end conditions are given by

$$\begin{aligned} \mathbf{v}(0) &= \mathbf{v}^0 = (x^0, y^0, z^0)^T \\ \mathbf{v}(1) &= \mathbf{v}^1 = (x^1, y^1, z^1)^T. \end{aligned} \tag{3}$$

Note that all the ϕ_is are the relevant costs appropriately weighted to be considered in the single objective function; and the C_ks are the various constraints to be imposed on the solution. For illustrative purpose, only earthwork cost ϕ_1 and pavement costs ϕ_2 are considered in the formulation:

$$\phi_1 = \alpha_1(\delta)A(\delta)\sqrt{(x'(t))^2 + (y'(t))^2} \tag{4a}$$

$$\phi_2 = \alpha_2\sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}, \tag{4b}$$

where

δ = $f(x(t), y(t)) - z(t)$ is the difference between the ground profile and road profile
$\alpha_1(\delta)$ = unit earthwork cost as a function of the depth of cut or fill
$A(\delta)$ = cross-sectional area of the cut or fill
α_2 = pavement cost per unit length.

Other costs that are quantifiable, like the users' cost, can easily be incorporated in the objective function if necessary. The constraints discussed previously can be explicitly considered as follows:

vertical slope

$$\left|\frac{dz}{ds}\right| \le C_1 \tag{5a}$$

vertical curvature

$$\left|\frac{\dfrac{d^2z}{ds^2}}{\sqrt{\left[1 + \left(\dfrac{dz}{ds}\right)^2\right]^3}}\right| \le C_2 \tag{5b}$$

horizontal curvature

$$\left|\frac{\dfrac{d^2y}{dx^2}}{\sqrt{\left[1 + \left(\dfrac{dy}{dx}\right)^2\right]^3}}\right| \le C_3 \tag{5c}$$

inaccessible region

$$\psi(x, y) \ge C_4, \tag{5d}$$

where

$$ds = \sqrt{dx^2 + dy^2} \tag{6}$$

is an infinitesimal axial element of the projection of road profile onto the x–y plane and C_1, C_2, and C_3 are constants representing various constraint controls.

The constraints have to be satisfied everywhere along the path. This becomes tedious computationally as there are infinite number of points in [0, 1]. To facilitate computation, the constraints are transformed into one-dimensional constraints via a constraint transcription used in optimal control theory (Teo and Goh, 1987; Goh and Teo, 1988). The proof is shown in Appendix B. The inequality constraints are now expressed as equality constraints:

slope constraint

$$g_1 = \eta_1 \int_0^1 \left\{ \min \left[C_1^2 - \frac{z'(t)^2}{x'(t)^2 + y'(t)^2}, 0 \right] \right\}^2 \mathrm{d}t = 0, \tag{7a}$$

vertical curvature constraint

$$g_2 = \eta^2 \int_0^1 \left\{ \min \left[C_2^2 - \left(\frac{z''(t)}{x'(t)^2 + y'(t)^2} - \frac{z'(t)(x'(t)x''(t) + y'(t)y''(t))}{(x'(t)^2 + y'(t)^2)^2} \right)^2, 0 \right] \right\}^2 \mathrm{d}t = 0, \tag{7b}$$

horizontal curvature constraint

$$g_3 = \eta_3 \int_0^1 \left\{ \min \left[C_3^2 - \frac{(x'(t)y''(t) - y'(t)x''(t))^2}{(x'(t)^2 + y'(t)^2)^3}, 0 \right] \right\}^2 \mathrm{d}t = 0, \tag{7c}$$

and inaccessible region constraint

$$g_4 = \eta_4 \int_0^1 \left\{ \min \left[C_4^2 - \Psi(x(t), y(t))^2, 0 \right] \right\}^2 \mathrm{d}t = 0, \tag{7d}$$

where the η_ks are empirical weighting factors that can be adjusted accordingly to enforce constraints satisfaction. They are used to improve the numerical accuracy and theoretically they can be taken as 1. However, in practice ηs may be varied depending on how far the intermediate solution is from the optimal solution.

To transform the above functional optimization problem into a mathematical programming problem, the states $x(t)$, $y(t)$, and $z(t)$ are parametrized as a series of cubic spline functions (Schumaker, 1969):

$$\mathbf{v(t)} = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} = \sum_{j=-1}^{N+1} \sigma_j \Omega(Nt - j), \tag{8}$$

where $\sigma_j = (\sigma_{1j}, \sigma_{2j}, \sigma_{3j})^T$ are the spline coefficients and Ω is the cubic spline function defined by

$$\Omega(\zeta) = \begin{cases} 0 & |\zeta| \geq 2 \\ \frac{1}{2}|\zeta|^3 - \zeta^2 + \frac{2}{3} & 0 \leq |\zeta| \leq 1 \\ -\frac{1}{6}|\zeta|^3 + \zeta^2 - 2|\zeta| + \frac{4}{3} & 1 \leq |\zeta| \leq 2. \end{cases} \tag{9}$$

Since the cubic spline functions $\Omega(\cdot)$ are of $C^{(2)}$, the parametric equations in (8) have continuous first and second derivatives. This is necessary as the curvature constraints

are required to be continuous since most standard optimization techniques require continuous gradient information and hence some degree of smoothness is needed to solve the problem.

The interval [0, 1] is partitioned into N equal intervals of width $\Delta = 1/N$ in each component of the vector $\mathbf{v}(\mathbf{t})$. In eqns (8) and (9), $x(t)$, $y(t)$, and $z(t)$ each comprises $N + 3$ cubic spline functions. Each spline function spans over a subdomain of width 4Δ centered at the points $-\Delta, 0, \Delta, 2\Delta, \ldots, N\Delta = 1, (N + 1)\Delta$. Thus, the vector $\mathbf{v}(\mathbf{t})$ that described the highway profile is represented by $3 \times (N + 3)$ spline coefficients σ_{ij}, $i = (1, 2, 3)$, and $j = -1, 0, 1, 2, \ldots, N + 1$. These coefficients are to be determined such that the cost function in eqn (4) is minimized subject to the constraints in eqn (7).

Note that the number of spline functions can be reduced to $N + 1$ in each component of vector $\mathbf{v}(\mathbf{t})$ by virtue of the fixed end boundary conditions given in eqn (3). It can easily be shown that

$$\sigma_{-1} = 6\mathbf{v}^0 - 4\sigma_0 - \sigma_1 \tag{10a}$$

and

$$\sigma_{N+1} = 6\mathbf{v}^1 - 4\sigma_N - \sigma_{N-1}, \tag{10b}$$

where $\mathbf{v}^0 = (x^0, y^0, z^0)^T$ and $\mathbf{v}^1 = (x^1, y^1, z^1)^T$ are the starting and end points, respectively. Hence, eqn (8) can be written in the following form:

$$\mathbf{v}(\mathbf{t}) = x(t), y(t), z(t))^T = (6\mathbf{v}^0 - 4\sigma_0 - \sigma_1)\Omega(Nt + 1) + (6\mathbf{v}^1 - 4\sigma_N - \sigma_{N-1})\Omega(Nt - N - 1) + \sum_{j=0}^{N} \sigma_j \Omega(Nt - j). \tag{11a}$$

The derivatives of $\mathbf{v}(\mathbf{t})$ can easily be shown to be

$$\mathbf{v}'(\mathbf{t}) = (x'(t), y'(t), z'(t))^T = (6\mathbf{v}^0 - 4\sigma_0 - \sigma_1)N\Omega'(Nt + 1) + (6\mathbf{v}^1 - 4\sigma_N - \sigma_{N-1})N\Omega'(Nt - N - 1) + \sum_{j=0}^{N} \sigma_j N\Omega'(Nt - j) \tag{11b}$$

and

$$\mathbf{v}''(\mathbf{t}) = (x''(t), y''(t), z''(t))^T = (6\mathbf{v}^0 - 4\sigma_0 - \sigma_1)N^2\Omega''(Nt + 1) + (6\mathbf{v}^1 - 4\sigma_N - \sigma_{N-1})N^2\Omega''(Nt - N - 1) + \sum_{j=0}^{N} \sigma_j N^2\Omega''(Nt - j). \tag{11c}$$

The vector $\mathbf{v}(\mathbf{t})$ that describes the road profile and its derivatives in eqn (11) are substituted into eqns (4) and (7) to yield the following constrained nonlinear programming problem:

$$\min_{\sigma} J(\sigma) \tag{12a}$$

subject to

$$g_k(\sigma) = 0 \quad k = 1, \ldots, m \tag{12b}$$

where

$$\sigma = (\sigma_{ij}, i = 1, 2, 3; j = 0, 1, \ldots, N). \tag{13}$$

The spline coefficients σ_{ij}s can be determined by using standard mathematical programming software such as the Generalized Reduced Gradient (Lasdon *et al.*, 1978) or NLPQL (Schittkowski, 1985) package. These softwares require gradient information about the cost and constraints functions. Summarized in Appendix A are the required partial derivatives with respect to spline coefficients.

Note that it is not possible to integrate the equations analytically. Numerical evaluation such as Simpson's rule or Romberg's adaptive interpolation scheme can be used to compute these integrals.

SCALING CONSIDERATION

The performance of the optimization algorithm is known to be critically dependent on the choice of variables used to define the problem (Luenberger, 1973). A bad choice may adversely affect the convergence characteristic. As the present problem involves a large number of variables, an appropriate scaling of these dimension variables is of paramount importance.

Let vector $\mathbf{r}$ be the original variables to be optimized represented by $(\bar{\sigma}_{1j}, \bar{\sigma}_{2j}, \bar{\sigma}_{3j})$; $\forall_j = 0, 1, \ldots, N$. The variables can be scaled such that the vector $\mathbf{r}$ is now changed to $\mathbf{v} = (\sigma_{1j}, \sigma_{2j}, \sigma_{3j})$ through a transformation matrix $\mathbf{T}$ given by

$$\mathbf{T} = \left| \begin{array}{c:c:c} \mathbf{T}_x & 0 & 0 \\ \hdashline 0 & \mathbf{T}_y & 0 \\ \hdashline 0 & 0 & \mathbf{T}_z \end{array} \right| , \tag{14}$$

where $\mathbf{T}_x$, $\mathbf{T}_y$, and $\mathbf{T}_z$ are block diagonal scaling factor matrices of the variables in direction x, y, and z, respectively. Hence, the transformation is

$$\mathbf{v} = \mathbf{T}^{-1}\mathbf{r}. \tag{15}$$

The problem of finding $\mathbf{r}$ to minimize $f(\mathbf{r})$ is equivalent to that of finding $\mathbf{v}$ to minimize $h(\mathbf{v}) = f(\mathbf{T}\mathbf{v})$. Using $\mathbf{v}$ as the underlying set of variables, we then have

$$\nabla h = \nabla f T. \tag{16}$$

In most quasi-Newton descend algorithms, the search direction can be represented in a general form

$$\mathbf{v}_{k+1} = \mathbf{v}_k - \alpha_k \mathbf{D}_k \nabla h^{\mathrm{T}}, \tag{17}$$

where

k = kth step of the algorithm used
α_k = positive step size parameter selected to satisfy certain descent conditions
D_k = $n \times n$ matrix approximating the inverse of the Hessian at $\mathbf{v}_k$.

From eqns (15), (16), and (17) it can easily be shown that

$$\Delta \mathbf{r}_k = -\alpha_k \mathbf{T}\mathbf{D}_k \mathbf{T}^{\mathrm{T}} \nabla f^{\mathrm{T}}, \tag{18}$$

We can see the scaling of variables changes the direction of the search. In the steepest descend method where matrix $\mathbf{D}$ is an identity matrix $\mathbf{I}$, the magnitude of the gradient term is increased by the square of the scaling factor. For example, if $\mathbf{T}_x = 100$, $\mathbf{T}_y = 100$, and $\mathbf{T}_z = 1$, we can observe that the gradients of the objective functions with

Table 1. Cost data and constraint conditions for text example

Item	Information	
Vertical gradient control	4% maximum AASHTO recommendation, 1984	
Vertical curvature	1,667 m (radius)	
Horizontal curvature	300 m (radius)	
Inaccessible region (2nd example)	Assumed a parabolic curve (Fig. 4a)	
Cutting costs	Depth of cut	Cost $/m³
	<1.5 m	S$10.00
	1.5–3.0 m	S$14.40
	3.0–4.5 m	S$18.20
	4.5–6.0 m	S$25.00
	6.0–7.5 m	S$30.00
	>7.5m	S$50.00
Filling costs	S$10.00/m³	
Pavement costs	S$50.00/m²	

respect to σ_{xj}s and σ_{yj}s are increased significantly. Hence, this gives rise to drastic changes of the variables in x and y directions but not the variables in z direction. This may lead to improper scaling and cause the variables in the x and y directions to reach near local optimums but not in the z direction, and they change slowly. This may require more iterations to complete the run.

Hence, proper scaling is necessary for obtaining a good solution. For a large-scale problem, there is no direct method of assigning proper scaling factors. Often, intuition and experience contribute greatly to a good choice.

NUMERICAL EXAMPLES

An actual ground profile in Singapore with an area of 2.2 × 1.6 square kilometers is considered. Based on the given profile, two alignment problems are solved. The first problem is to determine a highway alignment across a very irregular contours with steep slopes. The termini chosen are at coordinates (1100, 1100, 18.26) and (2800, 400, 15.24)

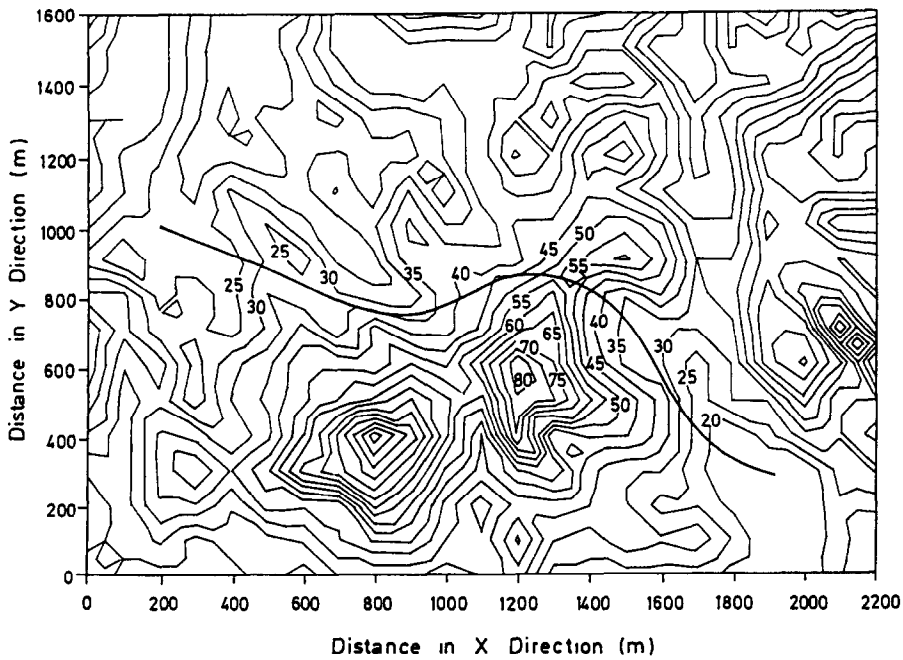

Fig. 3a. Optimal horizontal alignment for P1 by three-dimensional model.

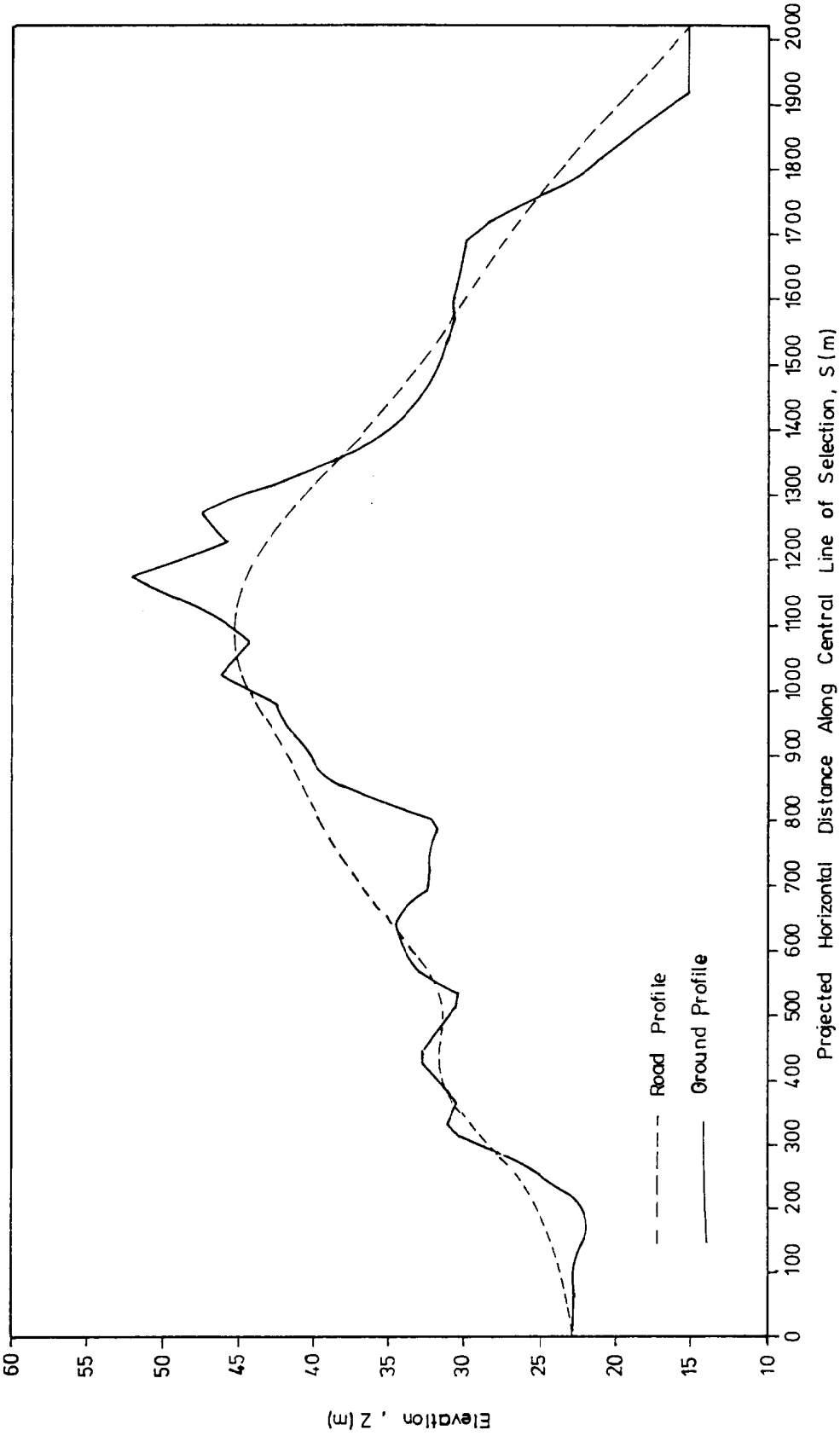

Fig. 3b. Optimal vertical alignment for P1 by three-dimensional model.

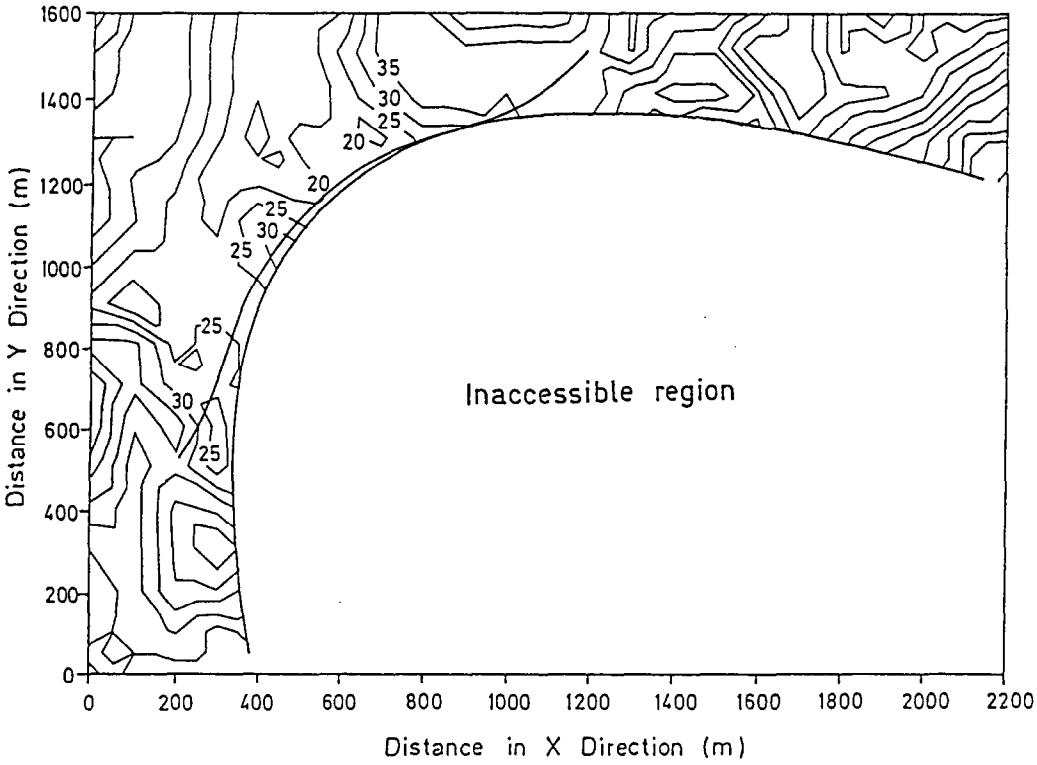

Fig. 4a. Optimal horizontal alignment for P2 by three-dimensional model.

where the units are measured in metres. The second problem is to align the highway through a terrain of gentler slope but with the inclusion of an inaccessible region. The coordinates of the termini are located at (1100, 600, 38.10) and (2100, 1600, 33.53), respectively.

The costs and constraint conditions are given in Table 1. The pavement costs in the examples are assumed to be constant over the region. Fifteen spline coefficients are spanned in each direction, thus resulting in a total of 45 optimization variables. This amount is chosen because after an increase from 36 to 45 variables, only about 1% improvement in the solution is obtained. Further increase would only increase the CPU processing time. For this amount of spline coefficient, each iteration search requires an average of three CPU minutes on the IBM 3081 when Generalized Reduced Gradient routine is used.

The computed horizontal and vertical alignments of the road profile for the first example problem are plotted in Figs. 3a and 3b, respectively. In Fig. 3a, the horizontal profile obtained tends to deviate from the steep contours so as to reduce earthwork costs. Instead of a path following a straight line joining the two end points, it chose to make a curve around the steep contours. Figure 3b shows the projection of the vertical alignment along the selected route. The optimal construction costs obtained was S$1.10 million (approximately U.S.$0.55 million).

The solution for the second problem is plotted in Figs. 4a and 4b. Since the region has much gentler slopes, earthwork and pavement costs did not vary much with changes of highway alignment. As a result, the presence of the inaccessible region forced the path to follow closely to its boundary. The optimal cost was found to be S$0.62 million (approximately U.S.$0.31 million).

PRACTICAL CONSIDERATION OF OPTIMIZATION TECHNIQUE

This section discusses optimization techniques for large systems, especially those systems that are three-dimensional in nature. An approach is recommended to help in determining the optimal solution within a feasible amount of processing time.

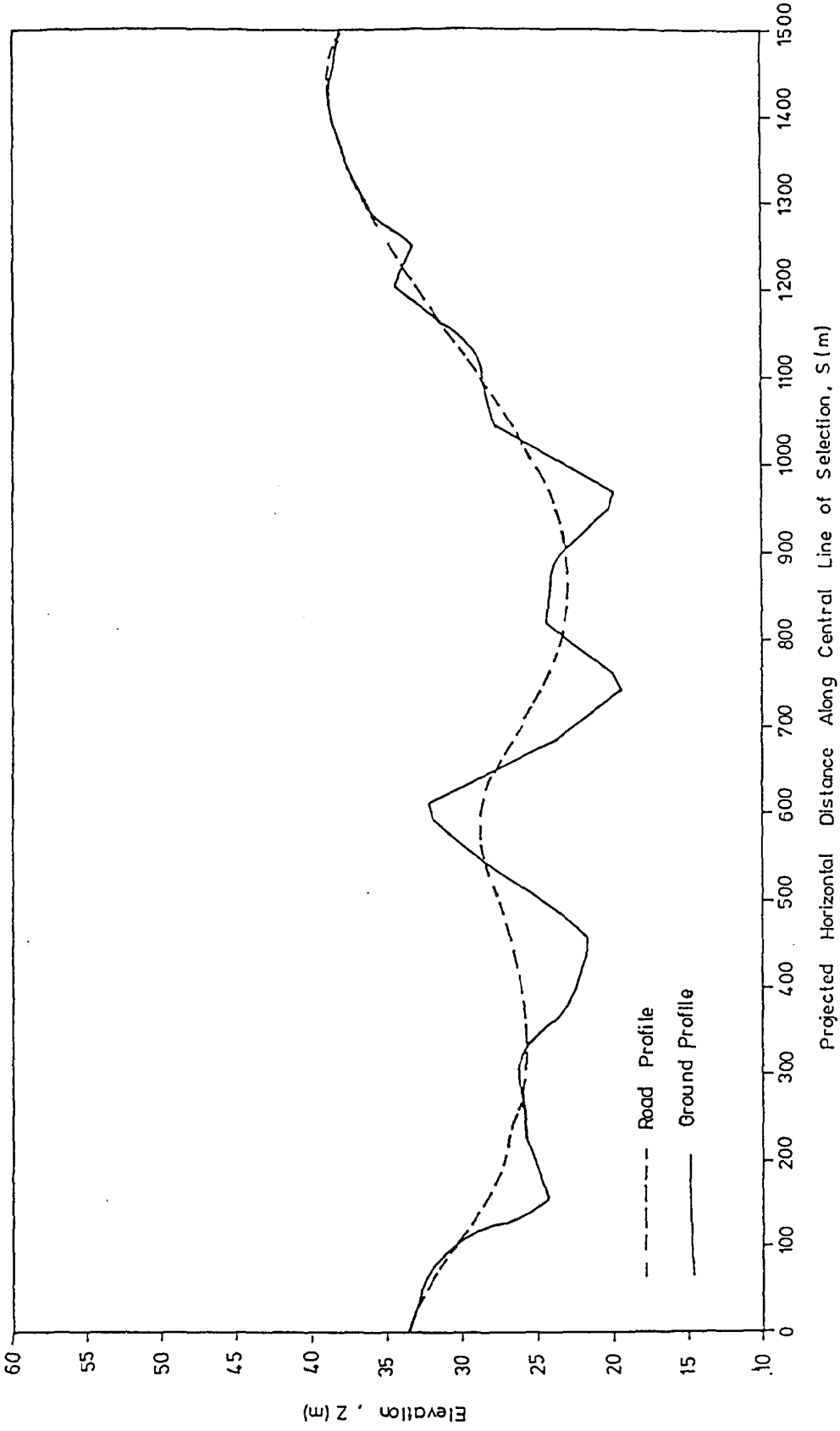

Fig. 4b. Optimal vertical alignment for P2 by three-dimensional model.

The selection of the initial variables is one of the important considerations as the rate of convergence depends on how close they are to the optimal solution. In selecting the variables, one has to study the contours in the region where the highway is to be constructed. Normally, one would tend to choose the variables such that the path will have the least earthwork activities. This is done by choosing the path that follows closely to the contour lines while observing the horizontal curvature constraint.

As this method might produce suboptimal solution, one strategy is to select several different feasible paths and choose the path that yield the minimum costs. In some optimization routines such as the Generalized Reduce Gradient, one can assume an infeasible path and the algorithm will start searching for a feasible path by incorporating it as part of the objective function. This value is reduced to the required tolerance that satisfies the constraint values and only then minimization of the objective function is carried out.

The rate of convergence is also dependent on the scaling of variables. The method recommended is to review the convergence rate after every p iterations where p is chosen to be five or six in this particular test problem. After each p iteration, the change of the variables is noted. The profiles of both the plan view and the vertical alignment are plotted to aid in the checking. If the horizontal alignment is found not to be sensitive to change when we suspect it should, then we can rescale the variables in the x and y direction such that the sensitivity of the gradient with respect to these variables is enhanced. Conversely, if the vertical alignment is not sensitive to change, then the z variables should be rescaled. This is done iteratively until the optimal solution is found or when scaling in the x, y, and z directions yields only negligible change.

CONCLUSIONS

A state parametrization model is developed to solve a three-dimensional highway alignment problem. Optimal vertical and horizontal alignments are determined simultaneously. Results from the two example problems are presented which show the capability of the model in solving practical situation. Any additional costs and constraints not included in the illustrated examples can be incorporated into the model as long as they can be represented by some forms of mathematical function.

However, one must also be aware that the choice of initial variables will affect the rate of convergence. For large problems with many variables, there is no analytical and direct means of selecting the initial values. One has to rely on intuition and accumulated experience. As an attempt to reduce the computation time, a technique of variable scaling is recommended in this paper.

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APPENDIX A: ANALYTICAL GRADIENT COMPUTATION

The partial derivatives of the objective function are:

$$\frac{\partial F}{\partial \sigma_{1,j}} = \int_0^1 \left\{ [\alpha_1'(\delta)A(\delta) + \alpha(\delta)A'(\delta)]\sqrt{x'^2 + y'^2}\, \frac{\partial \delta}{\partial x}\frac{\partial x}{\partial \sigma_{1,j}} + \left[\alpha_1(\delta)A(\delta)\frac{x'}{\sqrt{x'^2 + y'^2}} + \alpha_2 \frac{x'}{\sqrt{x'^2 + y'^2 + z'^2}}\right]\frac{\partial x'}{\partial \sigma_{1,j}}\right\} \mathrm{d}t \tag{19a}$$

$$\frac{\partial F}{\partial \sigma_{2,j}} = \int_0^1 \left\{ [\alpha_1'(\delta)A(\delta) + \alpha(\delta)A'(\delta)]\sqrt{x'^2 + y'^2}\, \frac{\partial \delta}{\partial y}\frac{\partial y}{\partial \sigma_{2,j}} + \left[\alpha_1(\delta)A(\delta)\frac{y'}{\sqrt{x'^2 + y'^2}} + \alpha_2 \frac{y'}{\sqrt{x'^2 + y'^2 + z'^2}}\right]\frac{\partial y'}{\partial \sigma_{2,j}}\right\} \mathrm{d}t \tag{19b}$$

$$\frac{\partial F}{\partial \sigma_{3,j}} = \int_0^1 \left\{ [\alpha_1'(\delta)A(\delta) + \alpha(\delta)A'(\delta)]\sqrt{x'^2 + y'^2}\, \frac{\partial \delta}{\partial z}\frac{\partial z}{\partial \sigma_{3,j}} + \alpha_2 \frac{z'}{\sqrt{x'^2 + y'^2 + z'^2}}\frac{\partial z'}{\partial \sigma_{3,j}}\right\} \mathrm{d}t, \tag{19c}$$

where

$$\alpha_1'(\delta) = \frac{\partial \alpha_1(\delta)}{\partial \delta} \text{ and } A'(\delta) = \frac{\partial A(\delta)}{\partial \delta}. \tag{19d}$$

The partial derivatives of the slope constraint are:

$$\frac{\partial g_1}{\partial \sigma_{1,j}} = 4\eta_1 \int_0^1 \left[\min\left(C_1^2 - \frac{z'^2}{x'^2 + y'^2}, 0\right)\right] \frac{z'^2 x'}{(x'^2 + y'^2)^2}\frac{\partial x'}{\partial \sigma_{1,j}}\, \mathrm{d}t \tag{20a}$$

$$\frac{\partial g_1}{\partial \sigma_{2,j}} = 4\eta_1 \int_0^1 \left[\min\left(C_1^2 - \frac{z'^2}{x'^2 + y'^2}, 0\right)\right] \frac{z'^2 y'}{(x'^2 + y'^2)^2}\frac{\partial y'}{\partial \sigma_{2,j}}\, \mathrm{d}t \tag{20b}$$

$$\frac{\partial g_1}{\partial \sigma_{3,j}} = -4\eta_1 \int_0^1 \left[\min\left(C_1^2 - \frac{z'^2}{x'^2 + y'^2}, 0\right)\right] \frac{z'}{(x'^2 + y'^2)}\frac{\partial z'}{\partial \sigma_{3,j}}\, \mathrm{d}t. \tag{20c}$$

The partial derivatives of the vertical curvature constraint are:

$$\frac{\partial g_2}{\partial \sigma_{1,j}} = 4\eta_2 \int_0^1 R_1 \left[\left(\frac{2z''x' + z'x''}{(x'^2 + y'^2)^2} - \frac{4x'z'(x'x'' + y'y'')}{(x'^2 + y'^2)^3}\right)\frac{\partial x'}{\partial \sigma_{1,j}} + \frac{z'x'}{(x'^2 + y'^2)^2}\frac{\partial x''}{\partial \sigma_{1,j}}\right) \mathrm{d}t \tag{21a}$$

$$\frac{\partial g_2}{\partial \sigma_{2,j}} = 4\eta_2 \int_0^1 R_1 \left[\left(\frac{2z''y' + z'y''}{(x'^2 + y'^2)^2} - \frac{4y'z'(x'x'' + y'y'')}{(x'^2 + y'^2)^3}\right)\frac{\partial y'}{\partial \sigma_{2,j}} + \frac{z'y'}{(x'^2 + y'^2)^2}\frac{\partial y''}{\partial \sigma_{2,j}}\right] \mathrm{d}t \tag{21b}$$

$$\frac{\partial g_2}{\partial \sigma_{3,j}} = 4\eta_2 \int_0^1 R_1 \left[\frac{(x'x'' + y'y'')}{(x'^2 + y'^2)^2}\frac{\partial z'}{\partial \sigma_{3,j}} - \frac{1}{x'^2 + y'^2}\frac{\partial z''}{\partial \sigma_{3,j}}\right] \mathrm{d}t. \tag{21c}$$

where

$$R_1 = \left[\min\left(C_2^2 - \left\{\frac{z''}{x'^2 + y'^2} - \frac{z'(x'x'' + y'y'')}{(x'^2 + y'^2)^2}\right\}^2, 0\right)\right]\left(\frac{z''}{x'^2 + y'^2} - \frac{z'(x'x'' + y'y'')}{(x'^2 + y'^2)^2}\right). \tag{21d}$$

The partial derivatives of the horizontal curvature constraint are:

$$\frac{\partial g_3}{\partial \sigma_{1,j}} = -4\eta_3 \int_0^1 R_2 \left[\left(\frac{y''(x'y'' - y'x'')}{(x'^2 + y'^2)^3} - \frac{3x'(x'y'' - y'x'')^2}{(x'^2 + y'^2)^4} \right) \frac{\partial x'}{\partial \sigma_{1,j}} - \frac{y'(x'y'' - y'x'')}{(x'^2 + y'^2)^3} \frac{\partial x''}{\partial \sigma_{1,j}} \right] \mathrm{d}t \tag{22a}$$

$$\frac{\partial g_3}{\partial \sigma_{2,j}} = 4\eta_3 \int_0^1 R_2 \left[\left(\frac{x''(x'y'' - y'x'')}{(x'^2 + y'^2)^3} + \frac{3y'(x'y'' - y'x'')^2}{(x'^2 + y'^2)^4} \right) \frac{\partial y'}{\partial \sigma_{2,j}} - \frac{x'(x'y'' - y'x'')}{(x'^2 + y'^2)^3} \frac{\partial y''}{\partial \sigma_{2,j}} \right] \mathrm{d}t \tag{22b}$$

$$\frac{\partial g_3}{\partial \sigma_{3,j}} = 0, \tag{22c}$$

where

$$R_2 = \min \left(C_3^2 - \left\{ \frac{(x'y'' - y'x'')^2}{(x'^2 + y'^2)^3} \right\}, 0 \right). \tag{22d}$$

The partial derivatives of the infeasible region are:

$$\frac{\partial g_4}{\partial \sigma_{1,j}} = -4\eta_4 \int_0^1 \min[C_4^2 - \Psi(x, y)^2, 0] \Psi \frac{\partial \Psi}{\partial x} \frac{\partial x}{\partial \sigma_{1,j}} \mathrm{d}t = 0 \tag{23a}$$

$$\frac{\partial g_4}{\partial \sigma_{2,j}} = -4\eta_4 \int_0^1 \min[C_4^2 - \Psi(x, y)^2, 0] \Psi \frac{\partial \Psi}{\partial y} \frac{\partial y}{\partial \sigma_{2,j}} \mathrm{d}t = 0 \tag{23b}$$

$$\frac{\partial g_4}{\partial \sigma_{3,j}} = 0 \qquad \forall j = 0, \ldots, N. \tag{23c}$$

APPENDIX B

The functional inequality constraints are given by

$$f_j(t) \geq 0 \quad j = 1, \ldots, m \quad \forall t \in [0, 1].$$

We would like to show that

$$f(t) \geq 0 \quad \forall t \in [0, 1] \quad \text{iff} \int_0^1 [\min(f(t), 0)]^2 \, \mathrm{d}t = 0.$$

Proof.
Sufficiency condition: if not, i.e.

$$\int_0^1 [\min(f(t), 0)]^2 \, \mathrm{d}t = \Delta > 0 \Rightarrow \exists \chi \ni f(t) < 0 \quad \forall t \in \chi,$$

which is a contradiction.
Necessary condition: if $\exists \Lambda \subset [0, 1] \ni f(t) < 0 \quad \forall t \in \Lambda$, it is clear again that

$$\int_\Lambda \{\min(f(t), 0)\}^2 \, \mathrm{d}t > 0 \Rightarrow \int_0^1 \{\min(f(t), 0)\}^2 \, \mathrm{d}t > 0,$$

which is a contradiction. Hence, the proof is complete. For further discussion, refer to Teo and Goh (1987). ■